PHY709: 15 October 2026

Maxwell equations, constitutive relations and electromagnetic boundary conditions

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Post-class problem set

Use SI units. Declare the interface normal before applying any jump condition. Show intermediate steps and verify each result against the starting equations.

1. A planar interface has n = z-hat, epsilon1 = 2 epsilon0 and epsilon2 = 4 epsilon0. Given E1 = 100 x-hat + 200 z-hat V/m and zero free surface charge, find E2.
2. At z = 0, n = z-hat, Kf = 6 x-hat A/m and H1 = 2 y-hat + 3 z-hat A/m. For equal permeabilities, find H2.
3. Derive all four electromagnetic boundary conditions. Identify the finite and vanishing terms in each pillbox or loop limit.
4. Derive the electrostatic potential equation for position-dependent scalar permittivity. Explain why a global Poisson equation with constant epsilon is insufficient.
5. Explain why tangential E remains continuous at an ordinary interface during electromagnetic induction. State the regularity assumption.

These are instructor-created exercises. They are not reproduced Jackson problems.